Homework 4: The Magnetosphere
Answer Key

Fall 2026

1 Field Magnitudes

Question Using this online calculator, determine the magnetic field in Boulder, CO, and at McMurdo Station, Antarctica. Report the total field magnitude, the declination, and the inclination. What does the inclination at McMurdo tell you about the magnetic field lines above this region?

Answer Here are the correct outputs for running IGRF or WMM from the provided link. The inclination at McMurdo is about -80°. This tells us that the magnetic field lines above this region are almost completely vertical; the location is pretty close to the magnetic pole.

Magnetic Field

Model Used:	IGRF2020
Latitude:	40° 0' 56" N
Longitude:	105° 16' 45" W
Elevation:	0.0 km Mean Sea Level

Date	Declination (+ E | - W)	Inclination (+ D | - U)	Horizontal Intensity	North Comp (+ N | - S)	East Comp (+ E | - W)	Vertical Comp (+ D | - U)	Total Field
2023-10-24	7° 50' 18"	66° 10' 39"	20,796.4 nT	20,602.1 nT	2,836.2 nT	47,102.0 nT	51,488.7 nT
Change/year	-0° 6' 18"/yr	-0° 2' 23"/yr	-9.9 nT/yr	-4.6 nT/yr	-39.1 nT/yr	-110.7 nT/yr	-105.3 nT/yr

Figure 1: IGRF model output for Boulder, CO.

Magnetic Field

Model Used:	IGRF2020
Latitude:	77° 50' 46" S
Longitude:	166° 40' 5" E
Elevation:	0.0 km Mean Sea Level

Date	Declination (+ E | - W)	Inclination (+ D | - U)	Horizontal Intensity	North Comp (+ N | - S)	East Comp (+ E | - W)	Vertical Comp (+ D | - U)	Total Field
2023-10-24	140° 43' 49"	-80° 12' 49"	10,559.9 nT	-8,175.2 nT	6,684.1 nT	-61,222.4 nT	62,126.4 nT
Change/year	-0° 11' 43"/yr	0° 2' 39"/yr	39.4 nT/yr	-7.7 nT/yr	52.8 nT/yr	53.3 nT/yr	-45.8 nT/yr

Figure 2: IGRF model output for McMurdo Station, Antarctica.

Magnetic Field

Model Used:	WMM-2020
Latitude:	40° 0' 56" N
Longitude:	105° 16' 45" W
Elevation:	0.0 km Mean Sea Level

Date	Declination (+ E | - W)	Inclination (+ D | - U)	Horizontal Intensity	North Comp (+ N | - S)	East Comp (+ E | - W)	Vertical Comp (+ D | - U)	Total Field
2023-10-24	7° 51' 28"	66° 10' 27"	20,799.7 nT	20,604.4 nT	2,843.6 nT	47,101.4 nT	51,489.5 nT
Change/year	-0° 5' 51"/yr	-0° 2' 28"/yr	-9.6 nT/yr	-4.6 nT/yr	-36.4 nT/yr	-113.3 nT/yr	-107.5 nT/yr
Uncertainty	0° 23'	0° 13'	128 nT	131 nT	94 nT	157 nT	145 nT

Figure 3: WMM model output for Boulder, CO.

Magnetic Field

Model Used:	WMM-2020
Latitude:	77° 50' 46" S
Longitude:	166° 40' 5" E
Elevation:	0.0 km Mean Sea Level

Date	Declination (+ E | - W)	Inclination (+ D | - U)	Horizontal Intensity	North Comp (+ N | - S)	East Comp (+ E | - W)	Vertical Comp (+ D | - U)	Total Field
2023-10-24	140° 49' 8"	-80° 12' 25"	10,565.3 nT	-8,189.7 nT	6,674.8 nT	-61,209.8 nT	62,115.0 nT
Change/year	-0° 11' 12"/yr	0° 2' 44"/yr	40.3 nT/yr	-9.5 nT/yr	52.2 nT/yr	55.8 nT/yr	-48.1 nT/yr
Uncertainty	0° 35'	0° 13'	128 nT	131 nT	94 nT	157 nT	145 nT

Figure 4: WMM model output for McMurdo Station, Antarctica.

2 Dipole Field

Question Assuming a dipole field model, what is the magnetic field magnitude at $L = 4$, (i) at the equator, and (ii) at $\lambda = 30°$?

Answer (i) The magnetic field magnitude at the equator ($\lambda = 0$) at $L = 4$ is given by:

$$\begin{aligned} |B| &= B_0 \left(\frac{R_E}{r}\right)^3 \sqrt{1 + 3\sin^2\lambda} \\ &= B_0 \left(\frac{R_E}{LR_E \cos^2\lambda}\right)^3 \sqrt{1 + 3\sin^2\lambda} \\ &= \frac{B_0}{L^3} = \frac{3.1 \times 10^{-5}\ \text{T}}{4^3} \simeq 4.8 \times 10^{-7}\ \text{T} = \boxed{480\ \text{nT}} \end{aligned}$$

(ii) Now we follow the procedure, but with $\lambda = 30°$:

$$\begin{aligned} |B| &= B_0 \left(\frac{R_E}{LR_E \cos^2\lambda}\right)^3 \sqrt{1 + 3\sin^2\lambda} \\ &= 3.1 \times 10^{-5}\ \text{T} \left(\frac{1}{4\cos^2(30°)}\right)^3 \sqrt{1 + 3\sin^2(30°)} \\ &\simeq 1.5 \times 10^{-6}\ \text{T} = \boxed{1500\ \text{nT}} \end{aligned}$$

Why is the field so much stronger at higher latitude? We know that the field is stronger towards the poles (compared to the equator) even at the same distance from the Earth. But more importantly, at higher latitude the point along that same field line is considerably closer to the Earth.

3 Magnetopause

Question The magnetopause is the boundary where the solar wind pressure equals the magnetic pressure of the magnetosphere.

(i) Assuming a dipole field with $B_0 = 31{,}000$ nT, and a solar wind moving at 400 km/s with a density of 5 protons/cm^3, estimate the standoff distance, i.e., the distance from the magnetopause from the center of the Earth.

(ii) During a particular CME, the solar wind increases to $v = 1000$ km/s and the proton number density doubles (these changes are typical, but it can vary!). What is the standoff distance under these conditions?

Answer (i) To estimate the magnetopause standoff distance at the equator, we set the solar wind ram pressure equal to the magnetic pressure of the magnetosphere:

$$(\rho v^2)_{SW} = \frac{4B(r)^2}{2\mu_0} = \frac{2(B_0/L^3)^2}{\mu_0} = \frac{2B_0^2}{\mu_0 L^6} \quad \rightarrow \quad L = \left(\frac{2B_0^2}{\mu_0 \rho v^2}\right)^{1/6}$$

$$L = \left[\frac{2(3.1\times10^{-5}\ \text{T})^2}{(1.256\times10^{-6}\ \text{N/A}^2)(5\times10^{6}\ \text{m}^{-3})(1.67\times10^{-27}\ \text{kg})(400\times10^{3}\ \text{m/s})^2}\right]^{1/6} = 10.2$$

Since we are above the equator, $L = 10.2$ is the same as $\boxed{10.2R_E.}$ This is our standoff distance!

(ii) If we solve again, but with $v = 1000$ km/s and $n = 10$ protons/cm^3, we find the standoff distance is at about $\boxed{6.7R_E}$.

4 The Ring Current

Question Ring current particles have an average energy of 85 keV. Consider the case where they have only $v_\perp$, i.e. $v_\parallel = 0$, and their motion is strictly in the equatorial plane. For this problem, consider the effects of the magnetic field gradient only.

(i) Explain why there is a current in the first place. (There are two key factors.)

(ii) How long does it take an 85 keV electron to drift all the way around the Earth at $L = 4$?

Answer (i) The ring current arises from ∇B (gradient) drift, though there is also a contribution from curvature drift. In the magnetosphere, ∇B is pointed toward the Earth and $\mathbf{B}$ is pointed up at the equator. With this configuration, the drift velocity for protons is westward and for electrons it is eastward. The net current westward is the ring current.

The two key factors are thus a) the motion of charged particles due to the gradient drift, and b) the fact that the drift direction is charge-dependent. If both electrons and ions drifted in the same direction (as in $E \times B$ drift), there would be no net current.

(ii) Our general approach is to solve for the ∇B drift velocity, v_d, find the total distance the particle will travel around the Earth, d, and use these values to solve for the drift time, $t = d/v_d$. ∇B drift velocity is:

$$\mathbf{v}_{d\nabla} = \frac{mv_\perp^2}{2q}\frac{\mathbf{B}\times\nabla B}{B^3}$$

Let's start by solving for ∇B, the gradient of the magnetic field. We know that the magnetic field at the equator is $\mathbf{B} = B_0(R_E/r)^3\hat{\boldsymbol{\theta}}$. Wikipedia's Del formula table is a good place for a quick reference if you forget (or never knew) the equations for divergence ($\nabla\cdot\mathbf{B}$), gradient (∇B), or curl ($\nabla\times\mathbf{B}$) in different coordinate systems.

$$\begin{aligned}
\nabla B &= \frac{\partial B}{\partial r}\hat{\mathbf{r}} + \frac{1}{r}\frac{\partial B}{\partial\theta}\hat{\boldsymbol{\theta}} + \frac{1}{r\sin\theta}\frac{\partial B}{\partial\varphi}\hat{\boldsymbol{\varphi}} \\
&= \frac{\partial}{\partial r}(B_0R_E^3r^{-3})\hat{\mathbf{r}} + \frac{1}{r}\cancelto{0}{\frac{\partial}{\partial\theta}(B_0R_E^3r^{-3})}\hat{\boldsymbol{\theta}} + \frac{1}{r\sin\theta}\cancelto{0}{\frac{\partial}{\partial\varphi}(B_0R_E^3r^{-3})}\hat{\boldsymbol{\varphi}} \\
&= B_0R_E^3\frac{\partial}{\partial r}(r^{-3})\hat{\mathbf{r}} \\
&= -3B_0R_E^3r^{-4}\hat{\mathbf{r}} \\
&= -3\frac{B}{r}\hat{\mathbf{r}}, \text{ where } B = B_0\frac{R_E^3}{r^3}
\end{aligned}$$

Next, let's find $\frac{\mathbf{B}\times\nabla B}{B^3}$:

$$\begin{aligned}
\frac{\mathbf{B}\times\nabla B}{B^3} &= \frac{B\hat{\boldsymbol{\theta}}\times(-3\frac{B}{r})\hat{\mathbf{r}}}{B^3} \\
&= \frac{-3B^2}{rB^3}(\hat{\boldsymbol{\theta}}\times\hat{\mathbf{r}}) \\
&= \frac{3}{r}\frac{1}{B}\hat{\boldsymbol{\phi}}
\end{aligned}$$

Now we are ready to solve for v_d:

$$\begin{aligned}
\mathbf{v_d} &= \frac{mv_\perp^2}{2q}\frac{\mathbf{B}\times\nabla B}{B^3} \\
&= \frac{E_{k\perp}}{q}\frac{3}{r}\frac{1}{B}\hat{\boldsymbol{\phi}} \\
&\simeq 20,700 \text{ m/s}
\end{aligned}$$

where $E_{k\perp} = mv_\perp^2/2$ is the perpendicular component of the kinetic energy of the particle.

To find the distance d the particle travels in one orbit, we just need to solve for the circumference of a circle with the radius equal to $4R_E$:

$$\begin{aligned}
d = 2\pi(4R_E) &= 2\pi(4\times6.371\times10^6 \text{ m}) \\
&\simeq 1.60\times10^8 \text{ m}
\end{aligned}$$

Finally, we can solve for time:

$$t = \frac{d}{v_d} = \frac{1.60\times10^8 \text{ m}}{20,700 \text{ m/s}} \simeq 7750 \text{ s} \simeq \boxed{2.15 \text{ hours}}$$

Note the form of the equation for v_d, written in terms of $E_{k\perp}$ and B. Higher energy particles have a faster drift speed, and thus a shorter drift time. But a *weaker* magnetic field (higher L-shell) leads to a faster drift speed as well. B varies as $1/r^3$, but distance varies as r, so higher L-shells end up with shorter drift periods, despite the increasingly larger distance the particle has to travel.

5 Drift due to Gravity

Question As above, consider charged particles near the equator at $L = 4$.

(i) Under the influence of gravity, what direction will protons and electrons drift?

(ii) What is the magnitude of the drift velocity due to gravity? (Don't forget that gravity is weaker as we move further from the Earth.)

(iii) How long will it take particles to drift around the Earth? Compare your answer to Problem 4 above.

Answer (i) Above the equator at $L = 4$, the gravity vector points towards the Earth and $\mathbf{B}$ points up. With this configuration, the gravitational drift velocity for protons is eastward and for electrons is westward.

(ii) The magnitude of the drift velocity due to gravity at the equator is:

$$\mathbf{v_g} = \frac{m}{q}\frac{\mathbf{g}\times\mathbf{B}}{B^2} \rightarrow |v_g| = \frac{m}{|q|}\frac{g}{B}$$

Note that the q in the original denominator determines that protons and electrons will drift in opposite directions, but the m in the numerator means electrons will have a gravitational drift velocity much less than protons. Before we solve for v_g, we need to find g at $L = 4$:

$$g = g_0\left(\frac{R_e}{R_e+h}\right)^2$$
$$g_{L=4} = 9.8\text{m/s}^2\left(\frac{1}{4}\right)^2 = 0.61\text{m/s}^2$$

We also need the value of B at $L = 4$ above the equator:

$$B = B_0\left(\frac{R_E}{r}\right)^3 = \frac{B_0}{L^3}$$
$$B_{L=4} = \frac{3.1\times10^{-5}\text{ T}}{4^3} = 4.84\times10^{-7}\text{ T}$$

For protons:

$$|v_g|_p = \frac{m_p}{|q_p|}\frac{g_{L=4}}{B_{L=4}} = \frac{1.67\times10^{-27}\text{ kg}}{1.602\times10^{-19}\text{ C}}\frac{0.61\text{ m/s}^2}{4.84\times10^{-7}\text{ T}} = \boxed{1.3\times10^{-2}\text{ m/s}}$$

For electrons:

$$|v_g|_e = \frac{m_e}{|q_e|}\frac{g_{L=4}}{B_{L=4}} = \frac{9.11\times10^{-31}\text{ kg}}{1.602\times10^{-19}\text{ C}}\frac{0.61\text{ m/s}^2}{4.84\times10^{-7}\text{ T}} = \boxed{7.2\times10^{-6}\text{ m/s}}$$

(iii) The gravitational drift time for one orbit for protons and electrons, respectively, is:

$$t_p = \frac{1.60 \times 10^8 \text{ m}}{1.3 \times 10^{-2} \text{ m/s}} \simeq 1.2 \times 10^{10} \text{ s} \simeq \boxed{380 \text{ years}}$$

$$t_e = \frac{1.60 \times 10^8 \text{ m}}{7.2 \times 10^{-6} \text{ m/s}} \simeq 2.2 \times 10^{13} \text{ s} \simeq \boxed{700 \text{ millennia}}$$

Where we found that gradient drift can move an 85 keV electron around the Earth at $L = 4$ in just over 2 hours, gravitational drift takes MUCH longer.

6 Geomagnetic Indicies

Question In this problem, we'll investigate the distribution of the Kp index over a 50-year period.

- go to https://omniweb.gsfc.nasa.gov/ow.html
- Click on "Distributions functions, averages and std deviations" in the top left yellow box.
- Click the radio button for "Get distribution function etc"
- Enter the start date as 19640101 and the end date as 20161231
- Click the radio button for Kp*10 Index (this will be the numerical version of Kp multiplied by 10)
- Further down on the page specify the bin size as 1
- Click Submit
- Include the figure in your assignment submission.

Now answer these questions:

i) What is the most likely value of Kp?

ii) What fraction of the Kp values is observed to be at level 9? (hint: click on "Listing of Distribution Function table" above the image.)

iii) Repeat the procedure above to get a histogram of the Kp data for the month of July 2004. Compare the relative frequency of the Kp > 5 for this interval to the multi-year interval above. Would you say that July 2004 was more or less disturbed than typical?

Note that you can use this online tool to determine the typical values and distribution functions of many solar wind parameters, energetic particles fluxes, geomagnetic indices, and more.

Answer Your plot should look like Figure 5 if your inputs are correct.

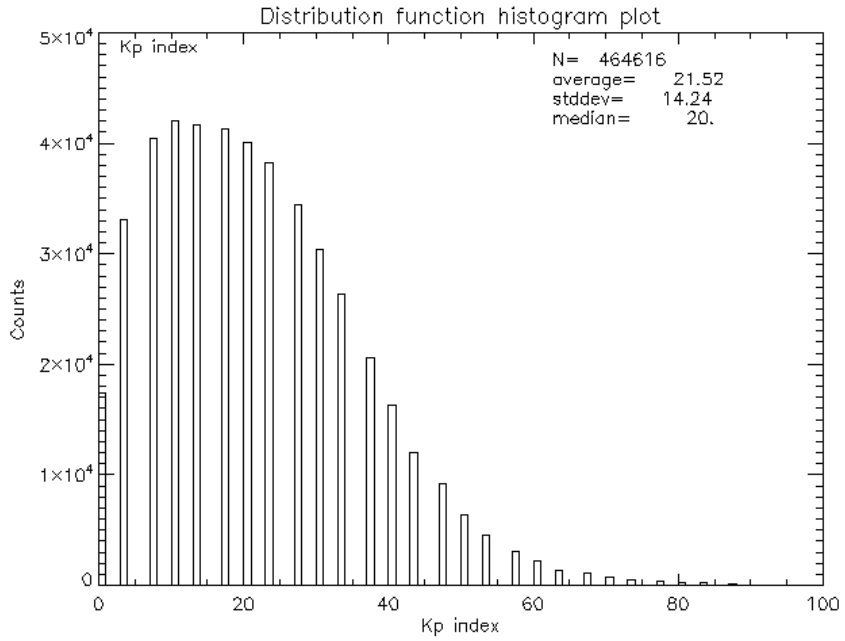


Figure 5: Kp index distribution.

i) The most likely value of Kp is 1 (Kp*10 = 10).

ii) 39.0 / 464616.0 ($\simeq 0.008\%$) of the Kp values are observed to be at level 9.

iii) The relative frequency of the Kp > 5 (which includes the values corresponding to Kp = 5+) for this interval is 87/744 ($\simeq 11.7\%$). The relative frequency of the Kp > 5 for the multi-year interval is 20907/464616 ($\simeq 4.49\%$). July 2004 was more disturbed than typical. Your exact percentages may vary depending on interpretation, but your conclusion should agree.

7 Geomagnetic Storm

Question In addition to OmniWeb above, historical Dst index data is available from this website: http://wdc.kugi.kyoto-u.ac.jp/dstdir/index.html.

(i) Use this site to create a figure showing the Dst index for the month of October 2003. (The event shown is known as the "Halloween Storm.")

(ii) Calculate the ring current magnitude (in Amperes) necessary to create the observed Dst of -400 nT. Assume the ring current can be modeled as a circular loop of current at $5R_E$.

Answer i) Navigate your way through the Kyoto site to get this figure:

ii) The solution to this problem is derived from the *Biot-Savart Law*, which you probably learned in introductory E&M, but also probably forgot. However, you would find this easily if you searched for "magnetic field at the center of a current loop". The Biot-Savart Law gives the differential magnetic field element, $d\mathbf{B}$, generated by a differential current element, $Id\vec{l}$:

$$d\mathbf{B} = \frac{\mu_0 Id\vec{l} \times \vec{r}}{4\pi r^2}$$



Figure 6: Dst (Final) for October 2003, taken from the WDC for Geomagnetism site.

where $\vec{r}$ points from the current element to the observation location, and r is the distance. We next integrate this equation over a distribution of current, such as a long wire or, in this case, a loop of wire with radius R. To do this most accurately at the surface of the earth, you need to integrate the equation above around the current loop, where r is changing throughout the integration (because you are observing **B** at a single location on the Earth's surface. But you'll get very close by simply evaluating **B** at the center of the current loop (i.e. at the center of the Earth). The magnetic field at the center dramatically simplifies to:

$$B = \frac{\mu_0 I}{2R}$$

(You could have just looked this equation up directly, and that would be fine!)

Rewriting and solving for the current, we find:

$$I = \frac{2BR}{\mu_0} = \frac{2 \times 400 \times 10^{-9} \times (5 \times 6371 \times 10^3)}{1.256 \times 10^{-6}} = 2.03 \times 10^7 \text{ Amperes} = \boxed{\sim 20.3 \text{ MA}}$$

8 Key Parameters in the Solar Wind Driving a Magnetic Storm

Question i) Among the parameters describing solar wind: velocity, plasma density, and interplanetary magnetic field, which parameter is the most critical one, in what way, to determine if a magnetic storm will occur? Why?

ii) Look at the Dst index profile for a magnetic storm (you should look through a few of them to convince yourself), you will see a certain surge (a positive change) in Dst index before the Dst index decreases (becomes more negative) as the main phase of the storm starts. What kind of current likely causes the initial surge of Dst index before the main phase of the magnetic storm?

Answer: (i) the the orientation and the magnitude of IMF is the most critical one. If there is sustain southward IMF with large amplitude, the IMF and Earth's day side magnetic field will reconnect, transporting more energy from solar wind into the magnetosphere, leading to the magnetic storm! (ii) The day side magnetopause current, which cancels the Earth's magnetic field outside the magnetopause and increase the Earth's magnetic field inside the magnetopasue

9 The Core

Question Just based on the 3-minute clip posted on Canvas, list (i) three statements about the space environment that are correct, and (ii) three statements about the space environment that are complete garbage.

Answer (i) Some true statements...

- *Invisible field of energy made up of electricity and magnetism*: Yes, especially the magnetic field, but it's true that there are also electric fields in the environment.
- *Protects us from cosmic radiation*: Yes, the magnetic field deflects and traps SEPs and GCRs (we'll see this more in the next module), though some of it reaches the atmosphere and even the surface (e.g. secondary particles produced by very high energy cosmic rays).
- Aaron Eckhart's description of the Earth's core: Pretty good description of the Earth's inner layers!
- *Physics 101, hot metal moving fast drives EM field*: this is actually generally correct; this is the basic principle behind the dynamo theory, which sounds simple but is somewhat complicated physics.

(ii) Some comically false statements...

- *Everybody on Earth is dead in a year.* Even if the magnetic field turned off instantly, the Earth's atmosphere would continue to protect us from radiation.
- It's not called the electromagnetic field; the field of interest is purely magnetic. We typically only use the term "electromagnetic" to refer to EM waves.
- *Planes will fall from the sky*: Why, exactly? They may not be able to navigate anymore (compasses would be useless), but compasses don't keep planes in the sky.
- *Static discharges in the atmosphere will lead to superstorms:* Static discharges in the atmosphere are also known as "lightning", and are a product of storms; they do not cause storms. Also they have very little to do with Earth's magnetic field.
- *Microwave radiation will literally cook our planet:* I guess this is why they focused on microwaves? Bonus points if anyone can quantitatively compare the field intensity (in V/m for the electric field) of microwaves i) from the sun and ii) in a microwave oven.

10 Magnetospheric Multiscale

(i) What is the primary objective of the MSS mission?

(ii) Why does the mission require four spacecraft? What arrangement of the four spacecraft is used?

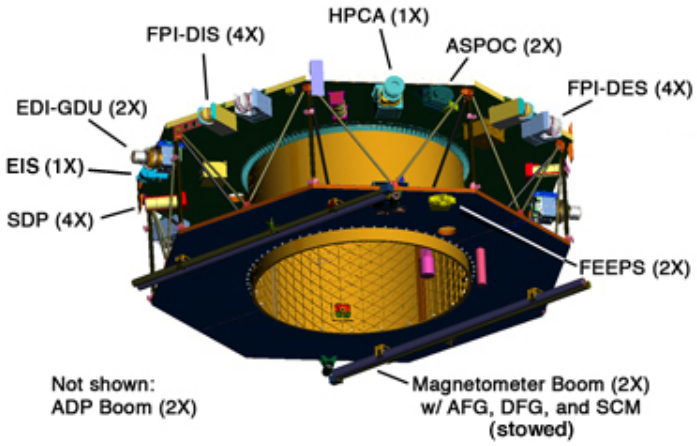
(iii) What instruments are on each of the four spacecraft? How many of each? Why are there multiples of some instruments? (hint: the spacecraft are *spin-stabilized*; what does that mean?)

(iv) What are the two main orbit phases of the MMS mission, and why? What are the apogee and perigee of each phase?

Answer (i) The overall science objective is to investigate the physics of magnetic reconnection. More specifically, the science objectives of the MMS mission are: (1) determine the role played by electron inertial effects and turbulent dissipation in driving reconnection in the electron diffusion region; (2) determine the rate of magnetic reconnection and the parameters that control it; (3) determine the role played by ion inertial effects in the physics of reconnection.

(ii) We need multiple spacecraft to see three-dimensional structure! Only by measuring the behavior of the particles and fields in three-dimensions in the diffusion region can we learn fully how reconnection proceeds. A tetrahedral formation is used; with this formation, if an event occurs within the tetrahedron, the source location can be triangulated in 3D from the four spacecraft measurements.

(iii) MMS's payload is comprised of three instrument groups: Hot Plasma, Energetic Particles, and Fields. In addition, the payload includes two ASPOC (Active Spacecraft Potential Control Devices) and a CIDP (Central Instrument Data Processor). The MMS spacecraft is spin-stabilized so that it is possible to determine full three-axis attitudes, instead of just the single spin-axis direction. The instruments and the total number of each instrument (per spacecraft) are shown below:



Instrument acronyms:
FPI = Fast Plasma Instrument
DIS = Dual Ion Sensors
DES = Dual Electron Sensors
HPCA = Hot Plasma Composition Analyzer
ASPOC = Active Spacecraft Potential Control Device
FEEPS = Fly's Eye Energetic Particle Sensor
EIS = Energetic Ion Spectrometer
EDI = Electron Drift Instrument
GDU = Gun Detector Unit
SDP = Spin-plane Double Probe
ADP = Axial Double Probe
AFG = Analog Fluxgate Magnetometer
DFG = Digital Fluxgate Magnetometer
SCM = Search Coil Magnetometer

(iv) Phase 1 (1.2 RE x 12 RE) was designed to probe reconnection sites at the mid-latitude dayside magnetopause, while Phase 2 (1.2 RE x 25 RE) was designed to focus on reconnection sites that occur within the nightside magnetic neutral current sheet.